PY-STEM · ACTIVITY PYS-08 · MECHANICS

Low-Ropes Force Map Relay

Predict balance and stability before the balance challenge proves you right.

Win condition: Most accurate predictions and clear debrief connections.

THE SCIENCE YOU ARE USING

Center of mass. Your center of mass is the average position of your weight. You stay balanced while it stays over your base of support. Lean too far and the line of gravity leaves the base, so you tip. Lowering your center of mass adds stability.

Mapping forces. At each balance challenge, you predict where balance will be hard, then test it with your body and a partner. The four challenges: (1) stand heels and back against a wall and try to lift your far foot or reach a coin on the floor ahead of you; (2) sit in a backless chair, torso straight, and try to stand without leaning forward; (3) bend, grab your toes, and try to hop forward; (4) walk a taped floor line holding a small weight at arm's length, then at your chest. Each one fails or works for a center-of-mass reason. Accurate predictions and a clear debrief score.

HOW TO BUILD AND RUN IT

01 **Read the challenge.** Look at the next balance challenge card and the body position it asks for.

02 **Predict balance points.** On the card, mark whether it will work or fail, and where your center of mass moves.

03 **Test it.** Do the challenge near the wall so you can catch yourself. Notice where you actually felt unstable.

04 **Map the forces.** Sketch where your weight acted and whether your center of mass stayed over your base of support.

05 **Debrief the connection.** Explain how lowering or shifting your center of mass changed stability.

Safety: Do every challenge next to the wall with your back and heels touching it so a slip just leaves you leaning on the wall. Keep the floor clear of bags and chairs. A helper spots the chair stand. The taped line is flat on the floor: no running, no raised beam.

MATERIALS

Element cards	set
Clipboards	6
Center-of-mass templates	per team
Pencils	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______

The ONE variable we are testing: ______

Baseline result: ______

After redesign: ______

DEFEND YOUR DESIGN

What evidence made you change your design? ______

If you ran it again, what is the ONE thing you would change? ______

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	**100**